

Problems on Logic

1. Define a proposition and truth value of a proposition
2. Identify the proposition among the sentences given below. Also write the truth values for the propositions.
 - (i) $a(b - c) = ab - ac$, for a, b, c being real
 - (ii) $\sqrt{3}$ is an irrational number
 - (iii) Square of every real number is positive
 - (iv) Sum of any two irrational numbers is irrational
 - (v) Force is a vector
 - (vi) The sum $3 + 2 \geq 4$.
 - (vii) Mathematics is a difficult subject
 - (viii) A quadrilateral has 4 sides and 4 vertices
 - (ix) A politician says that all politicians are liars
 - (x) $\sqrt{3}$ divides $2\sqrt{3}$
 - (xi) Vasanth finds logic difficult
 - (xii) Today is a wonderful day
 - (xiii) All squares are triangles
 - (xiv) Do your duty
 - (xv) How is your friend?
3. Which of the following is not a proposition? (CET 2003)
 - (i) 5 is an even integer
 - (ii) Mathematics is interesting
 - (iii) $\sqrt{2}$ is irrational
 - (iv) 3 is a prime number.

1. Define the compound propositions
 - (i) $p \wedge q$
 - (ii) $p \vee q$
 - (iii) $\sim p$
 - (iv) $p \rightarrow q$
 - (v) $p \leftrightarrow q$
2. Write the following compound propositions symbolically
 - (i) Varun is intelligent and Rajiv is a sportsman.
 - (ii) Mathematics is easy for Anu or science is not difficult for her.
 - (iii) If the question paper is difficult and I get good marks, then I am happy.
 - (iv) $(a + b)^2 = a^2 + b^2$ iff $ab + ba = 0$ ($a, b, 0$ are 2×2 square matrices)
 - (v) If $3^2 + 4^2 = 5^2$, then 4 is an irrational number.
 - (vi) If 2 is a composite number, then $3 + 5 = 7$ or $3^2 = 5^2$
 - (vii) He is hard working but does not get good marks.
 - (viii) It is either raining, or not raining and pleasant.

- (ix) If $a^2 = b^2$, then $a = b$ or $a = -b$
- (x) If a quadrilateral is a rhombus, then its diagonals are perpendicular. (This is the same as : the diagonals of a rhombus are perpendicular).
- (xi) Real numbers are complex numbers and, $x > 7$ or $x < 7$.
- (xii) If I have to study in NIT - K, I need to write AIEEE and if I have to get a management seat in a private medical college, I need to write COMED - K.
- (xiii) AIEEE is harder than JEE, JEE is harder than CET and CET is harder than PU Board Exam.
3. Given the propositions : p : I am clever ; q : Mathematics is interesting for me, r : I like logic.
Express in sentences each of the following compound propositions.
(i) $p \wedge q$ (ii) $p \rightarrow (q \wedge r)$ (iii) $r \rightarrow (p \vee q)$ (iv) $q \leftrightarrow r$ (v) $\sim p \wedge q$ (vi) $\sim p \wedge \sim r$.
4. Express the following compound propositions in words, taking p for "he worked hard" q for "he was successful in the examination" r for "he got a job".
(i) $(p \wedge q) \wedge \sim r$ (ii) $\sim p \wedge \sim q$ (iii) $p \wedge \sim q$ (iv) $(\sim p \wedge \sim q) \wedge r$ (v) $\sim (p \vee q) \wedge r$ (vi) $p \rightarrow q$ (vii) $(\sim q) \rightarrow (\sim p)$
5. If p : Ashok win; q : Babu is happy ; r : Chandra wins ; s : Dravid is happy
express the following compound propositions in words.
(i) $p \rightarrow q$ (ii) $r \rightarrow s$ (iii) $p \vee r$ (iv) $p \rightarrow \sim s$ (v) $r \rightarrow \sim q$ (vi) $q \leftrightarrow \sim s$
6. If p : 6 is an even number, q : 4 is a prime number and r : $6 + 4 = 10$, write down the truth values of
(i) $p \vee q$ (ii) $\sim p \wedge q$ (iii) $p \rightarrow \sim (q \wedge r)$ (iv) $\sim p \leftrightarrow (\sim p \vee r)$ (v) $r \leftrightarrow (p \wedge q)$
(vi) $\sim q \rightarrow (\sim p \wedge r)$ (vii) $p \leftrightarrow (q \rightarrow r)$
7. Determine the truth values of the following propositions
(i) $2 + 3 = 5$ and $4 \times 3 = 6$ (ii) $2 \times 3 = 6$ or $2 \times 3 = 7$ (iii) If $4 + 3 = 5$, then $3 + 4 = 5$
(iv) If $2 + 2 = 5$ and $3 + 3 = 6$, then $5 + 5 = 10$
8. Suppose p, q, r are propositions with truth values T, F, T respectively. Determine the truth values of the following
(i) $p \vee q$ (ii) $\sim q \wedge r$ (iii) $(p \vee q) \wedge \sim r$ (iv) $(p \wedge q) \wedge \sim r$ (v) $p \rightarrow (\sim q \vee r)$ (vi) $q \rightarrow (\sim p \rightarrow r)$
(vii) $p \leftrightarrow (q \rightarrow r)$ (viii) $p \wedge (q \rightarrow r)$ (ix) $\sim p \vee (\sim q \rightarrow \sim r)$ (x) $(\sim p \rightarrow \sim q) \rightarrow (\sim r \rightarrow p)$
9. Given that the proposition
(i) $p \wedge q$ is true and q is false. Find the truth value of p
(ii) $(p \wedge q) \rightarrow r$ is false. Find the truth values of p, q, r .
(iii) $p \rightarrow (q \vee r)$ is false. Find the truth values of p, q, r (CET 2000)
(iv) $p \leftrightarrow q$ is false and p is false. Find the truth value of q .
(v) $p \rightarrow (\sim q \vee \sim r)$ is false. Find the truth values of p, q, r .
(vi) $\sim q \wedge (\sim p \wedge r)$ is true. Find the truth values of p, q, r .
(vii) $(p \rightarrow q) \vee (p \rightarrow r)$ is false. Find the truth values of p, q, r .
(viii) $(\sim p \wedge q) \rightarrow (r \vee s)$ is false. Find the truth values of $p \rightarrow s$.
(ix) $(p \wedge q) \wedge (\sim r \wedge s)$ is true. Find the truth values of $(p \vee r) \rightarrow (q \vee s)$
(x) $p \rightarrow (q \wedge r)$ is false. Find the truth value of $q \rightarrow (p \rightarrow r)$.

10. Construct the truth tables for the following propositions

- (i) $p \wedge \sim q$ (ii) $\sim(p \wedge q)$ (iii) $(p \wedge q) \rightarrow (p \vee q)$ (iv) $\sim(p \rightarrow q)$ (v) $\sim p \rightarrow (q \wedge r)$
 (vi) $p \vee (q \wedge r)$ (vii) $(p \vee q) \wedge (p \wedge r)$ (viii) $(p \wedge q) \vee (\sim p \rightarrow q)$ (ix) $(q \rightarrow p) \vee (\sim p \leftrightarrow q)$
 (x) $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$

1. Define a tautology and a contradiction.

2. Prove that the following compound propositions are tautologies.

- (i) $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$ (ii) $[(p \vee q) \wedge (\sim p)] \rightarrow q$ (iii) $\sim(p \wedge q) \vee (q \rightarrow p)$
 (iv) $[p \rightarrow (q \wedge r)] \rightarrow [(p \rightarrow q) \wedge (p \rightarrow r)]$ (v) $[p \vee (q \wedge r)] \leftrightarrow [(p \vee q) \wedge (p \vee r)]$ (MQP)

3. Prove that the following compound propositions are contradictions

- (i) $(p \wedge q) \wedge (\sim p)$ (ii) $\sim p \wedge [(p \vee \sim q) \wedge q]$ (iii) $(p \wedge q) \wedge \sim(p \vee q)$ (iv) $[(p \rightarrow q) \rightarrow p] \wedge \sim p$
 (v) $(p \rightarrow \sim p) \wedge (\sim p \rightarrow p)$ (CET 97)

4. Prove that each of the following compound propositions is neither a tautology nor a contradiction

- (i) $p \vee (p \wedge q)$ (ii) $(p \rightarrow q) \wedge (q \rightarrow p)$ (iii) $(\sim p \wedge q) \wedge p$ (iv) $[(p \wedge q) \vee (p \vee q)] \rightarrow \sim q$

5. Examine whether the following compound propositions are tautologies, contradictions or neither.

- (i) $p \wedge (q \vee r) \leftrightarrow [(p \wedge q) \vee (p \wedge r)]$ (ii) $(p \wedge q) \wedge \sim(p \vee q)$ (iii) $[(p \rightarrow q) \wedge (q \rightarrow \sim r)] \rightarrow (p \rightarrow r)$
 (iv) $(p \vee \sim p) \wedge p$ (v) $\sim(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$ (vi) $(p \vee q) \leftrightarrow (\sim p \wedge \sim q)$

6. (i) Identify the false statement (CET 2002)

- (a) $p \wedge \sim p$ is a contradiction (b) $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$ is a contradiction
 (c) $\sim(\sim p) \leftrightarrow p$ is a tautology (d) $p \vee (\sim p)$ is tautology.

7. Define logically equivalent propositions.

8. Show that the following propositions are logically equivalent.

- (i) $p \wedge q \equiv q \wedge p$ (ii) $(p \wedge q) \equiv \sim(p \rightarrow \sim q)$ and $(\sim p \wedge q) \equiv \sim(q \rightarrow p)$ (CET 04)
 (iii) $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$; $p \vee (q \vee r) \equiv (p \vee q) \vee r$
 (iv) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$; $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
 (v) $\sim(p \vee q) \equiv \sim p \wedge \sim q$ (vi) $\sim(p \wedge q) \equiv \sim p \vee \sim q$
 (vii) $\sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (\sim p \wedge q)$ (viii) $\sim(\sim p \rightarrow \sim q) \equiv \sim p \wedge q$
 (ix) $[p \rightarrow (q \wedge r)] \equiv [\sim p \vee \sim(q \vee \sim r)]$ (x) $[\sim(p \vee q) \vee (\sim p \wedge q)] \equiv [(\sim p \wedge \sim q) \vee (\sim p \wedge q)]$

9. Which of the following is always true? (CET 98)

- (a) $p \rightarrow q \equiv (\sim q \rightarrow \sim p)$ (b) $\sim(p \vee q) \equiv (\sim p \vee \sim q)$ (c) $\sim(p \rightarrow q) \equiv (p \vee \sim q)$ (d) $\sim(p \wedge q) \equiv \sim p \wedge \sim q$

10. Using the logical equivalences of negations of compound propositions, negate the following propositions in symbols

- (i) $p \rightarrow \sim q$ (ii) $\sim p \wedge \sim q$ (iii) $p \rightarrow (q \rightarrow r)$ (iv) $p \rightarrow (q \vee r)$ (v) $p \leftrightarrow \sim q$ (vi) $(p \vee q) \rightarrow (q \wedge r)$
 (vii) $\sim p \leftrightarrow (p \wedge \sim q)$ (viii) $(p \rightarrow q) \leftrightarrow (p \vee r)$ (ix) $[(p \vee (q \wedge r)) \rightarrow p]$ (x) $\sim(p \vee q) \rightarrow (\sim p \wedge \sim q)$

11. Negate the following propositions in sentences

- (i) Laxmi is intelligent and hard working.
 (ii) Shwag scored a triple century in a test match but not Sachin Tendulkar.
 (iii) If cubes of two real numbers are equal, then they are equal.
 (iv) If 5 divides 7, then 4 divides 61.
 (v) If he does not score 60%, he cannot go for engineering.

- (vi) If a number is real, then it is rational or irrational.
- (vii) If it rains heavily, the college is closed but the students do not go home.
- (viii) If two triangles are not similar, then they are not equiangular.
- (ix) I will get a seat in Kasturba Medical College if and only if I am intelligent and rich.
- (x) I will get a merit seat if and only if I am hard working.

12. Define converse, inverse and contrapositive of a conditional and show that

- (i) a conditional and its contrapositive are logically equivalent.
- (ii) the converse and the inverse of a conditional are logically equivalent.

13. State the converse, inverse and the contrapositive of the following conditionals

- (i) If $a = b + c$, then $a^2 = b^2 + c^2$ (a, b, c are integers)
- (ii) If a quadrilateral is a rectangle, then it is a parallelogram
- (iii) If two real numbers are unequal, then their squares are also unequal
- (iv) If n is a prime number, then it is odd (CET 04)
- (v) If $a > b$, then $a < b$ (a and b are reals)
- (vi) If I work hard, I succeed
- (vii) If a triangle is a square, then $5 = 3$
- (viii) $p \rightarrow \sim q$
- (ix) $(p \wedge q) \rightarrow r$
- (x) $p \rightarrow (q \rightarrow r)$

14. (i) Write the converse of the contrapositive of $p \rightarrow q$ (CET 05)

(ii) Write the contrapositive of the inverse of $p \rightarrow \sim q$ (CET 05)

(iii) Write the contrapositive of "If two triangles are $\cong$, then these are similar" (CET 05)

1. Draw a switching circuit for the given Boolean polynomial.

(i) $(A \wedge B') \vee (A' \vee B)$ (ii) $p \vee (p' \wedge q)$ (iii) $(A \vee B) \wedge [A' \vee (C \vee B')]$

(iv) $(p' \vee r') \wedge (q' \vee p) \wedge (r \wedge q)$ (v) $(p \wedge q') \vee (p \wedge q) \vee (p' \wedge q')$

2. Find a Boolean expression for each of the following switching circuit.

